SURVEYING

Handwritten Study Notes

Subject :	SURVEYING
Code :	23A01301T
Unit :	III – Theodolite Surveying & Traversing

1. Theodolite – Introduction

A ***theodolite*** is the most accurate surveying instrument for measuring ***horizontal and vertical angles*** . It can also measure distances (tacheometry), prolong a line, and establish levels.

- Least count: 10, 20 or 30 seconds (vernier); 1 second for optical micrometre types.
- Uses: traversing, triangulation, setting out curves, tacheometric surveys, setting out buildings, heights of towers.

2. Types of Theodolites

Basis	Types
Transit / Non-transit	***Transit:*** telescope can be revolved through a complete vertical circle about its horizontal axis (commonly used). ***Non-transit:*** telescope cannot be transited (old).
Reading system	Vernier theodolite; micrometer (optical) theodolite; ***electronic / digital theodolite*** (reads angles by a display).
Size	Measured by the diameter of the lower plate: 8 cm, 10 cm, 12 cm, 15 cm and 25 cm.
Repeating / Directional	***Repeating:*** upper and lower plates can be clamped independently; for the repetition method. ***Directional (reiterating):*** only one plate; for the reiteration method.

Parts of a vernier transit theodolite

- ***Levelling head:*** tribrach with three foot screws; trivet; clamp to the tripod.
- ***Lower (scale) plate*** with the horizontal graduated circle; lower clamp and lower

tangent screw.

- ***Upper (vernier) plate*** with two verniers at 180 deg apart; upper clamp and upper tangent screw; plate levels.
- ***Standards (A-frame)*** support the ***telescope*** and the ***vertical circle*** (graduated), with verniers C and D on the T-frame (index arm) and a clip screw.
- ***Telescope*** with objective, eyepiece, diaphragm, focusing screw; ***altitude bubble*** ; plumb bob hook; compass (optional).

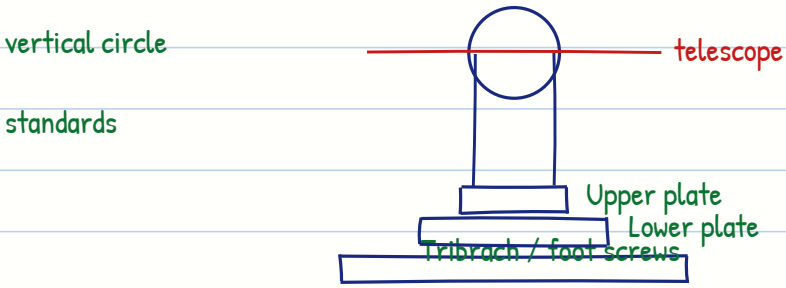


Fig 1: sketch of a transit theodolite

Technical terms

- ***Centring:*** keeping the vertical axis of the instrument exactly over the station mark.
- ***Transiting (plunging):*** revolving the telescope in a vertical plane through 180 deg about the horizontal axis.
- ***Face left / face right:*** vertical circle on left (face left) or on the right of the telescope (face right) when observing.
- ***Swinging the telescope:*** turning it in the horizontal plane. Right swing = clockwise; left swing = anticlockwise.
- ***Line of collimation:*** line through the cross hairs and the optical centre of the objective.
- ***Axis of the telescope, horizontal axis (trunnion axis), vertical axis*** - three axes that should be mutually at right angles.
- ***Telescope normal:*** vertical circle on the left, bubble of the telescope up. ***Inverted:*** vertical circle on the right, bubble down.

3. Temporary Adjustments of a Theodolite

Temporary adjustments are done at every station before taking observations:

- **Setting up:** place the tripod over the station with the head about level; fix the instrument; centre approximately by plumb bob and bring the plumb bob over the mark.
- **Centring:** shift the instrument on the tripod head (with the shifting head) until the plumb bob is exactly over the station.
- **Levelling up:** with the foot screws, bring the plate bubbles to the centre. Turn the upper plate until the plate level is parallel to two foot screws; centre the bubble; turn it 90 deg and centre using the third screw. Repeat till the bubble remains central in all positions.
- **Elimination of parallax:** (a) focus the eyepiece for distinct cross hairs, (b) focus the objective to see the object clear without any apparent movement of the cross hairs.

> **SEQUENCE:**
> Order: Setting up -> Centring -> Levelling -> Focusing the eyepiece and the objective. Permanent adjustments are done by the manufacturer / in the lab.

Permanent adjustments (relationship of axes)

- Plate bubble axis perpendicular to the vertical axis.
- Line of collimation perpendicular to the horizontal axis.
- Horizontal axis perpendicular to the vertical axis.
- Altitude bubble axis parallel to the line of collimation; vertical circle vernier reads zero when the line of sight is horizontal (index error nil).

4. Measurement of a Horizontal Angle

Ordinary method (single angle)

- Set up at station O and level. Set A vernier to zero (or any reading) using the upper plate; clamp the lower plate; sight A; fix with the lower clamp and tangent screw.
- Loosen the upper clamp; turn the telescope to sight B; clamp; bisect B with the tangent screw. Read both verniers; the mean difference from the initial reading gives angle AOB.

- Change face and repeat; the mean of the face-left and face-right values eliminates collimation and other errors.

Repetition method

Used to measure a horizontal angle **accurately** by **repeating** it several times and adding up on the circle. The final reading divided by the number of repetitions gives the angle. It reduces the least count error and the error of the instrument's graduation.

- Step 1: Measure angle AOB in the usual way with the upper clamp loose, lower clamp tight. Note the reading (e.g. a1).
- Step 2: Loosen the **lower** clamp, sight A again (reading stays the same), clamp lower; loosen upper, sight B. Reading is now a2 = twice the angle.
- Step 3: Repeat 3 times on face left, then change face and repeat 3 times on face right (total 6 repetitions).

Angle = (final reading – initial reading + 360 x number of full rotations) / number of repetitions

Least count error is reduced by n (i.e. 1/n of the original)

NOTE:

Repetition method needs a repeating (two-plate) theodolite. Errors due to poor graduation and eccentricity are eliminated; it cannot remove errors from slip, sighting, centring.

Example of observation booking

Station	To	Face	Vernier A	Vernier B	Mean	Angle
O	A	L	0 00 00	180 00 20	0 00 10	
	B	L	62 30 40	242 31 00	62 30 50	62 30 40
O	A	R	0 00 00	180 00 20	0 00 10	
	B	R	62 30 20	242 30 40	62 30 30	62 30 20

Mean angle = (62 30 40 + 62 30 20)/2 = 62 deg 30' 30".

Reiteration method

Used when **several angles at a station** are to be measured (angles AOB, BOC, COD, DOA). It is suited to a directional theodolite and avoids the accumulation of error in

measuring each separate angle.

- Set the instrument at O, level and centre. Sight the starting (reference) object A with a reading close to zero (first vernier 0 deg 00 00), the lower plate clamped. Read both verniers.
- Unclamp the upper plate, sight B, read; then C, D etc. in clockwise order, and finally close on A again (reading must agree with the initial reading within the least count – this is the closing check).
- Change face and repeat in the anticlockwise order (D, C, B, A). Take the mean of the two sets.
- To reduce errors, repeat in several ***sets*** with a different initial reading each time: 0, 60, 120 ... degrees; take the mean of corresponding angles.

5. Measurement of Vertical Angle

A ***vertical angle*** is the angle between the line of sight and the horizontal plane, measured in a vertical plane. It is an angle of elevation (+) or depression (–).

- Set up and level the theodolite at the station; bring the altitude bubble to the centre (the vertical circle verniers read zero when the line of sight is horizontal).
- Sight the object (use the horizontal cross hair); clamp the telescope; set the altitude bubble central by the clip screw; read both vertical verniers C and D.
- Change face and repeat; the mean of the readings eliminates the ***index error*** .

Vertical angle = (reading with the line of sight horizontal) – (reading at the object)

Index error = [(Reading face left) + (Reading face right) – 360 deg] / 2 ... (when read from 0 to 360)

Errors that cancel by changing face

- Error of collimation and horizontal axis; error of verticality of the vertical axis does not cancel.
- Eccentricity of verniers (by reading both verniers) and errors in graduation; index error of the vertical circle.

6. Trigonometrical Levelling

Trigonometrical levelling determines the difference in elevation of stations by the ***vertical angles*** and the horizontal distance (computed or measured). It is used in hilly regions, for heights of towers, chimneys, and for contour control.

(A) Base of the object is accessible

The instrument is set at station P; the horizontal distance D from P to the foot of the object Q can be measured directly. Let alpha = angle of elevation to the top, h = height of the instrument axis above the ground (= staff reading on B.M.).

Height of object H = D tan alpha (above the instrument axis)

R.L. of top = R.L. of P + h + D tan alpha

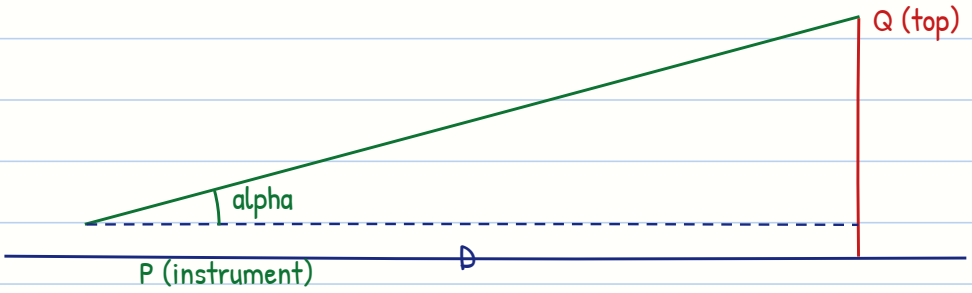


Fig 2: accessible base

(B) Base is inaccessible - instrument stations in the same vertical plane as th

The foot of the object is not accessible; two instrument stations P and Q are set in line with the object, a known distance b apart. Angles of elevation alpha1 at P (nearer) and alpha2 at Q (farther) are measured; the staff readings at B.M. are S1 and S2.

Case 1 (stations at the same level):

H = b tan(alpha1) tan(alpha2) / [tan(alpha1) - tan(alpha2)] = D1 tan alpha1

where D1 = b tan(alpha2) / [tan(alpha1) - tan(alpha2)]

Here P is nearer to the object with angle alpha1 > alpha2. D1 is the horizontal distance from P to the object. Then R.L. of object = R.L. of P (axis) + H.

Case 2 (axes of the two stations at different levels): find the height of the top

separately above each axis (using the staff readings S1 and S2 on the B.M. to get the R.L. of each axis); the two values of R.L. of the top should agree and the mean is taken.

(C) Base inaccessible – instrument stations NOT in the same vertical plane as

Two stations P and Q, distance b apart, are chosen such that P, Q and the object are not in a line. Horizontal angles at P and Q to the object are measured (angle alpha at P, angle beta at Q), and the vertical angles.

In triangle PQR: angle at R = 180 – (alpha + beta)

PR = b sin(beta) / sin(alpha + beta) (sine rule)

Height H = PR tan(vertical angle at P)

CORRECTION:

In trigonometrical levelling over long distances apply curvature and refraction correction: C = 0.0673 d^2 (m) to the computed height (add to the elevation of the far station).

Example

From a station P, the angle of elevation to the top of a chimney is 24 deg 15'. The base is accessible with a horizontal distance 85 m. The height of the instrument is 1.45 m.

H = 85 x tan 24 deg 15' = 85 x 0.4505 = 38.29 m

Total height above P ground = 38.29 + 1.45 = 39.74 m

7. Traversing

A ***traverse*** is a series of connected survey lines whose lengths and directions (angles or bearings) are measured. The survey is called ***traversing*** .

- ***Closed traverse:*** starts and ends at the same point (closed loop), or at two points of known position. Provides checks – used for boundary, plots, and control.
- ***Open traverse:*** does not close – used for long linear works like roads, canals, pipelines; has no checks.

Methods of traversing

Method	Description
Chain traversing	Angles by chain (triangles and tie lines); only for small areas

Chain and compass traversing	Bearings by compass; distances by chain. Also called free or loose needle method
Transit tape traversing	Angles by theodolite and distances by tape; for accurate work
Plane table traversing	Direction of lines by alidade and plot as the survey goes on
Theodolite traversing	Angles by theodolite; distances by tape or tacheometer or EDM

Theodolite traversing - methods of measuring angles

- ***By measuring included (interior) angles:*** angle between adjacent lines, usually in the anticlockwise direction for closed polygon (interior angles). Suits closed traverse, easily checked: sum = (2n - 4) x 90 deg.
- ***By direct observation of angles to the right (deflection angles):*** angle between the prolongation of the preceding line and the next line. Positive right, negative left. For open traverse - roads. Check: right deflection - left deflection = +/- 360 deg.
- ***By measuring bearings (magnetic) of lines:*** directly with the compass or with the theodolite fitted with a trough compass - loose needle method (angle from magnetic meridian at each station).
- ***Fast needle method (direct observation of bearing):*** the bearing of the first line is observed and then the angles; the bearings of other lines are computed.

8. Traverse Computations

Consecutive coordinates - Latitude and Departure

The ***latitude (L)*** of a line is its projection on the north-south axis (meridian); the ***departure (D)*** is its projection on the east-west axis.

Latitude $L = l \cos(\theta)$ (N = +, S = -)

Departure $D = l \sin(\theta)$ (E = +, W = -)

l = length of line; theta = W.C.B. of the line

W.C.B. (quadrant)	Latitude	Departure
0-90 (NE)	+	+

90-180 (SE)	-	+
180-270 (SW)	-	-
270-360 (NW)	+	-

Closing error

For a closed traverse, the algebraic sum of latitudes (Sum L) and of departures (Sum D) should each be zero. The non-zero sums give the **closing error** .

Closing error e = sqrt[(Sum L)^2 + (Sum D)^2]

Direction of error: tan(theta) = Sum D / Sum L

Relative precision = e / (perimeter) = 1 / (P / e)

Angular adjustment first

- Sum of interior angles of a closed polygon = (2n - 4) x 90 deg. The angular error = observed sum - theoretical sum.
- Distribute the angular error equally among all the angles (if equal weights) or in proportion to the weights.
- Recompute the bearings of the lines from the corrected angles; then compute L and D.

Balancing the traverse (adjusting the closing error)

Bowditch's rule (compass rule): used where angular and linear measurements are of the same accuracy. Correction in a line is proportional to its length.

Correction to latitude of a line = - (Sum L) x (l / Sum l)

Correction to departure of a line = - (Sum D) x (l / Sum l)

Transit rule: used when angular measurement is more precise than linear measurement (theodolite with taped distances).

Correction to latitude = - (Sum L) x (|L| / Sum |L|)

Correction to departure = - (Sum D) x (|D| / Sum |D|)

REMEMBER:
Sum l = perimeter. Sum |L| and Sum |D| = sum of the latitudes / departures without regard to the sign. Compass (Bowditch) rule is the most widely used.

Graphical method of adjusting (Bowditch)

Plot the traverse to scale as ABCDE (not closed, A and E' apart). Join A to E' and draw lines parallel to it, from B, C, D. The closing error is distributed along the traverse in proportion to the length of the sides (a straight line of cumulative lengths).

9. Gale's Traverse Table

A systematic tabular method of computing a closed traverse - includes the angles, bearings, lengths, latitudes, departures, corrections, adjusted values and the independent coordinates of each station.

Col.	Entry
1-2	Station; observed interior angle; correction; corrected angle
3	W.C.B. (from the preceding bearing + 180 - interior angle, etc.)
4	Length of line
5-6	Latitude and departure (+ and - columns)
7-8	Corrections by Bowditch; balanced latitudes and departures
9	Independent coordinates (N, E) of the stations (Northing = previous + L)

- Independent coordinates: Northing N = N(prev) + latitude; Easting E = E(prev) + departure.
- Choose the origin so that all the stations have positive coordinates (the most south-westerly station as origin).
- Area of the traverse can be determined from the coordinates (see Unit II).

10. Worked Example - Traverse Computation

A closed traverse ABCDA: AB = 100 m (N 0), BC = 80 m (E 90), CD = 100 m (S 180) and DA = 79 m (W 270). Check: Sum L = 100 - 100 = 0; Sum D = 80 - 79 = +1.

Closing error e = 1.0 m; perimeter = 359 m
Relative precision = 1/359

Bowditch: correction to departure of each line = -(+1.00) x l / 359. So AB = -0.28, BC

= -0.22, CD = -0.28, DA = -0.22; total = -1.00 m, which balances the +1.00 m error. Latitudes need no correction here (Sum L = 0).

II. Omitted Measurements (Introduction)

Sometimes the length or the bearing (or both) of one or two lines of a closed traverse cannot be measured in the field. These ***omitted measurements*** are computed from the condition that the traverse closes: Sum L = 0 and Sum D = 0.

Omitted	Method of finding
Length of one line	Sum of the known L and D gives the missing L and D; length = sqrt(L^2 + D^2); bearing = tan^-1(D / L)
Bearing of one line	Closing line of all other lines; cosine / sine rule (triangle formed by the missing line and the closing line)
Length and bearing of one line	Closing line of the rest is the missing line - closing error of the rest of the traverse is reversed
Length of one line and bearing of another	Solve a triangle by the sine rule with the closing line of the known sides
Lengths of two lines / Bearing of two lines	Solve using the closing line and the triangle (two unknowns, two equations)

- General idea: assume the known lines add up to a closing line (Sum L, Sum D); the missing portion makes up the closure.

Missing line: Latitude = - Sum L (known lines); Departure = - Sum D (known lines)

Length = sqrt(L^2 + D^2); Bearing = tan^-1 (D / L), with the proper quadrant

LIMIT:

All the missing measurements of one line (length and bearing) can be found by this method. If more than two quantities are omitted, the traverse cannot be solved by itself.

12. Sources of Errors in Theodolite Work

- **Instrumental:** imperfect adjustments, eccentricity of the verniers and the circle, graduation errors; eliminated by face-left/face-right and reading both verniers.
- **Personal:** inexact centring, bisection errors, poor levelling, parallax, wrong reading or booking.
- **Natural:** unequal refraction, wind, temperature, settling of tripod, sun on one side of instrument.
- Remedies: shade the instrument, tighten the tripod legs, measure on both faces, repeat the angles and check the sum.

13. Quick Revision

- Theodolite angles: repetition (single angle, accurate), reiteration (many angles at a station).
- Interior angles sum = (2n - 4) x 90 deg; Latitude = l cos(theta); Departure = l sin(theta).
- Bowditch for equal accuracy; transit rule when angles are more precise.
- Trigonometrical levelling: accessible base H = D tan(alpha); inaccessible base - two-station method.
- Omitted measurements: use closure Sum L = 0, Sum D = 0.